

Recommendation Systems for Personalized Content Discovery

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Problem Overview

- **Objective**

- Build a recommendation engine using the Netflix Prize Dataset.
- Learn user preferences from historical ratings.
- Predict ratings for unseen movies.
- Generate personalized Top-K movie recommendations.

- **Dataset**

- 100M+ ratings
- 480K+ users
- 17K+ movies
- Ratings: 1–5 stars
- Includes timestamps and movie metadata

Recommendation Approaches

Item Based
Collaborative
Filtering

SVD Matrix
Factorization

Key Insights

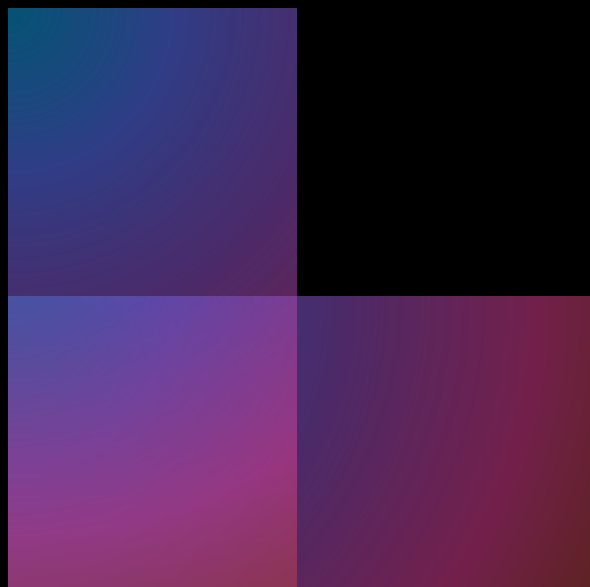
Highly Sparse Dataset

Long-Tail Movie Popularity

Matrix Factorization Outperformed
Traditional Similarity Methods

Prediction Accuracy $\neq$
Recommendation Quality

Ranking (MAP) and Error (RMSE)
Don't Always Scale Linearly



Experimental Results

- There is no absolute single winner, rather, both models excel in different production tiers.
- But for core matrix prediction accuracy, system scalability, and lightning-fast real-time inference at scale, **SVD Matrix Factorization** emerges as our primary **production-ready recommendation model**.

Metric	Item Based Model	SVD Model
RMSE	0.9431	0.821
MAP@10	5.65%	5.68%

Recommendation Examples

```
=====
Recommendation Generation using both models and Analysis
=====
Enter the value of K (Number of recommendations to generate): 10

Processing Top-10 Recommendations for a randomly chosen user...

User_ID: 1327900
Total Unseen Movies available for this user in Test Set: 101
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=====
TOP-10 MOVIE RECOMMENDATIONS BY BOTH MODELS
=====
```

ITEM-BASED MODEL			SVD-MODEL		
Item_Movie_ID	Item_Pred	Item_Actual	SVD_Movie_ID	SVD_Pred	SVD_Actual
2178	4.33	5.0	2102	4.82	5.0
4185	4.20	4.0	1202	4.58	4.0
3182	4.13	4.0	2452	4.56	5.0
571	4.10	5.0	571	4.50	5.0
3223	4.10	4.0	2122	4.44	5.0
1466	4.10	3.0	634	4.43	5.0
2102	4.08	5.0	1865	4.37	4.0
3900	4.08	3.0	3282	4.36	5.0
1865	4.08	4.0	3864	4.30	4.0
1408	4.08	3.0	2178	4.26	5.0

```
=====
```

MODEL 1: ITEM-BASED MODEL

Success Rate : 70.0%

Failure Rate : 30.0%

MODEL 2: SVD MATRIX FACTORIZATION MODEL

Success Rate : 100.0%

Failure Rate : 0.0%

Both the models have recommended 10 Movies to a user based on the rating predictions they have made .

A successful prediction is one where actual rating ≥ 3.5

Observation:
SVD has more success rate than Item Based Model

Conclusion

- Developed a personalized movie recommendation system using the Netflix Prize Dataset.
- Performed extensive analysis of user behavior, rating patterns, and data sparsity.
- Implemented and compared multiple recommendation approaches.
- Generated Top-10 personalized movie recommendations for users.
- Evaluated performance using RMSE and MAP@10 to assess both prediction accuracy and ranking quality.

